Deadline-Feasible, Uncertainty-Aware Budget Allocation for Shared Open RAN Fronthaul Links

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***Abstract*—In the Open RAN 7-2x split several radio units share one packet-switched fronthaul link that is smaller than the sum of their peak rates, and the distributed unit must decide, once per control interval and before the traffic of that interval is known, how much of the link each cell may use. We cast this as a constrained multi-item newsvendor problem whose random demand is the *deadline-feasible rate* of each cell, the smallest constant rate at which no bit of the coming interval misses its deadline. A gradient-boosted quantile forecaster supplies the predictive distribution, and the KKT conditions give a closed-form allocation that cuts every cell at the same priority-weighted tail probability, so safety margins follow from forecast uncertainty instead of a tuned global margin. We prove feasibility, equal weighted risk, backlog priority and a robustness bound under forecast miscalibration. On a slot-level simulator with two independent traffic families, a bursty synthetic family with a held-out shift and a standards-based family (3GPP TR 38.901 channel, TS 38.214 link adaptation, FTP-3 and XR traffic), the rule is the best implementable controller in every scenario and reduces the priority-weighted deadline-violation ratio by 11–43 % and 11–15 % against the same controller driven by the forecast median. We also report where the rule stops helping: the gain is a priority trade that needs a low-latency weight of at least two to three, and it vanishes when the control interval or the telemetry delay is far longer than the shortest deadline.**

***Index Terms*—Open RAN, fronthaul, functional split 7-2x, resource allocation, quantile forecasting, newsvendor, deadline-aware scheduling**

I. Introduction

In the O-RAN 7-2x functional split [1], [2] the radio unit (O-RU) sends frequency-domain IQ samples to the distributed unit (O-DU) for every scheduled physical resource block (PRB). The rate is several Gbit/s per cell and, unlike the constant-rate splits 7-1 and 8, follows the scheduled load [3], so operators aggregate several O-RUs onto one Ethernet link smaller than the sum of the cell peaks [4], [5]. When bursts coincide some cells have to wait, and low-latency services tolerate only a few milliseconds of waiting.

Besides block floating point (BFP) compression [1], the operator's lever is to decide every few milliseconds how much of the link each cell may use. The difficulty is that the decision is made in advance: a budget is a reservation held by the O-DU scheduler or a switch policer for a whole control interval, the telemetry behind it is a few slots old, and the arrivals of the coming interval are unknown. Prior work on shared 7-2x fronthaul controls compression reactively, slot by slot [6]. Forecast-driven capacity allocation [7]–[10] does plan ahead, but it works from a single predicted number, often inflated by a tuned margin, and it does not model per-cell deadlines. Probabilistic forecasts have been used [11], [12], with the percentile chosen globally or per stream.

Our question is this: if each cell's need for the next interval is available as a forecast *distribution*, how should a shared link be split between cells with different deadlines and priorities, and does the answer matter against strong reactive and point-forecast controllers? We make three contributions.

1) We forecast the deadline-feasible rate r^* of each cell rather than its volume. For a 2-ms class r^* is essentially the peak 2-ms burst rate and is strongly right-skewed, which is why a point forecast is a poor basis for a reservation (Sec. IV).
2) We show that maximising priority-weighted expected served demand under the forecast distributions, subject to the link constraint, is a separable concave program whose KKT conditions give the closed form $r_i = B_i + F_i^{-1}(1 - \lambda/w_i)$ with a single scalar multiplier (Sec. V). We prove feasibility, equal weighted risk, backlog priority and a robustness bound: a forecast with Kolmogorov error ϵ loses at most $2\epsilon\, w_{\max} C$.
3) We compare twelve controllers, among them MaxWeight, a tuned global-quantile rule, point-forecast and non-learned ablations and two oracle references, on identical traces with identical stale observations, on a bursty synthetic family with a held-out shift and on a standards-based 3GPP family, with 20 seeds per scenario and Holm-corrected Wilcoxon statistics (Secs. VI and VII; Fig. 1). We also report where the method loses, including under an unweighted objective.

II. Related Work

On fronthaul sharing, Larsen et al. [3] establish that the 7-2 bit rate follows the user load, Pérez et al. [4] dimension packet-switched eCPRI aggregation by delay percentiles, and Wang and Zhou [5] quantify the multiplexing gain. The deployment closest to ours is that of Lagén et al. [6], several 7-2x

cells behind an L2 switch with per-slot reactive compression control; we keep their deployment and study what happens when the control loop is slower than a slot. Public O-RAN datasets such as ColO-RAN [18] report KPIs at 250-ms or coarser resolution, which is why Sec. VI has to rely on models.

On forecast-driven allocation, DeepCog and AZTEC [7], [8] forecast slice demand with a loss that encodes the cost of over- versus under-provisioning, and [9], [10] feed point forecasts to slice admission or fronthaul-constrained allocation. None of them models per-cell deadlines, and the risk trade-off is fixed inside the model or by a margin. Closer to us, Kasuluru et al. [11] forecast PRB load probabilistically but pick one percentile globally, and Cohen et al. [12] apply adaptive conformal inference [13] to a single URLLC stream. Our rule instead derives a different quantile for every cell from a coupled allocation, and our ablation shows that per-stream conformal correction does not carry over to it.

Finally, MaxWeight and drift-plus-penalty [14] are the natural comparators that use no forecast. Our rule is a constrained multi-item newsvendor [15], and the deadline-feasible rate comes from network-calculus service curves [16].

III. System Model

We consider the downlink of N 7-2x cells that connect through a fronthaul switch to co-located O-DUs over one Ethernet link of capacity C_{link}. Time is slotted with $\Delta = 0.5$ ms (numerology 1). A fraction ρ_c of the link is reserved for the control and management planes, so the user-plane capacity is $C = (1-\rho_c)C_{\text{link}}\Delta$ bits per slot. Unless stated otherwise, $N = 8$, $C_{\text{link}} = 25$ Gbit/s, $\rho_c = 3\,\%$ and $C = 12.125$ Mbit/slot.

With BFP compression of b mantissa bits, a shared 8-bit exponent per PRB, 14 symbols per slot and an overhead factor κ, one PRB-layer for one slot costs

$$\beta(b) = (12 \cdot 2 \cdot b + 8) \cdot 14 \cdot \kappa \text{ bits}, \tag{1}$$

which is 3387 bits for $b = 9$ and $\kappa = 1.08$. A 273-PRB cell with $L = 4$ layers therefore peaks at $\text{cap}_i = 273L\beta$ bits per slot (7.4 Gbit/s), and eight such cells peak at $2.44\,C$. If cell i has to deliver $a_i(t)$ user bits in slot t at $\eta_i(t)$ user bits per PRB-layer, carrying them immediately needs

$$d_i(t) = \lceil a_i(t)/\eta_i(t) \rceil \beta \tag{2}$$

fronthaul bits; fronthaul load follows scheduled PRB-layers, not application bits [3].

Each cell keeps a FIFO queue in the O-DU of user bits waiting for fronthaul capacity, and belongs to a class with deadline D_i slots and operator priority w_i^{p}. By default there are three low-latency (LL) cells with $D = 4$ slots (2 ms) and $w^{\text{p}} = 3$, and five eMBB cells with $D = 20$ slots (10 ms) and $w^{\text{p}} = 1$. Bits that have waited D_i slots are discarded and counted as deadline violations. Given a budget of r_i bits per slot, the cell schedules $\lfloor r_i/\beta \rfloor$ PRB-layers, bounded by its air-interface capacity, oldest bits first. Because the budgets keep the link from queueing, switch queues are not modelled.

Every T slots (20 by default, i.e. 10 ms) the controller sets budgets r with

$$\textstyle\sum_i r_i \le C, \qquad 0 \le r_i \le \text{cap}_i, \tag{3}$$

and holds them for the interval. It sees queue lengths, head-of-line ages, spectral efficiencies and the demand history up to slot $t - \tau$ (by default $\tau = 2$ slots), and never the future arrivals, so a decision has to cover $H = T + \tau$ slots.

Our primary metric, fixed before any tuning, is the priority-weighted deadline-violation ratio $V_w = \sum_i w_i^{\text{p}} \text{violated}_i \big/ \sum_i w_i^{\text{p}} \text{arrived}_i$ over post-warm-up slots. It is the objective of an operator who pays w^{p} per lost LL bit relative to an eMBB bit. Because this choice matters, Sec. VII-C also reports the unweighted ratio ($w^{\text{p}} \equiv 1$) and a sweep of w_{LL}^{p}; per-class ratios, utilisation and decision time are reported as well.

IV. Deadline-Feasible Demand

We first have to say what a cell actually needs. Let $a = (a_1, \ldots, a_H)$ be the fronthaul bits arriving in the H slots of a horizon and D the deadline. The deadline-feasible rate is the smallest constant rate at which no bit waits longer than D slots,

$$r^*(a, D) = \max_{1 \le s \le e \le H} \frac{\sum_{t=s}^{e} a_t}{e - s + 1 + D}. \tag{4}$$

This is the network-calculus condition for a rate-latency server to meet the arrival curve [16]; it is tight, and it tends to the mean rate for $D \gg H$ and to the peak $(D{+}1)$-slot window rate for $D \ll H$. With $D = 4$ and $H = 22$ the LL value is set by the single strongest 2-ms burst and is strongly right-skewed across horizons, whereas the eMBB value stays close to the interval mean. The budget a cell needs is then

$$\text{need}_i = B_i + r_i^*, \tag{5}$$

where the backlog rate B_i clears the observed backlog within the slack left before its oldest bits expire, and r_i^* accounts for the arrivals not yet seen. All deadline-aware controllers below use (5); they differ in how r_i^* is estimated and in how a shortage is split between cells.

V. Proposed Allocation Rule

The rule has two parts, a forecast of r_i^* as a distribution and an allocation that turns these distributions into budgets. The forecaster provides quantiles $q_i(\alpha_k)$ of r_i^* for the coming horizon at the levels 0.05, 0.25, 0.5, 0.75, 0.9 and 0.95. We train one histogram gradient-boosted tree model per quantile (pinball loss, 300 trees) on 28 features that use past slots only: demand lags, exponentially weighted means at three horizons, the last horizon's r^* and mean, the recent 90th percentile of r^*, queue length, head-of-line age, spectral efficiency and its trend, and a cell one-hot; targets are normalised by the cell peak. Quantiles are sorted after prediction, and F_i^{-1} is the piecewise-linear interpolation through the knots, extrapolated linearly above 0.95. The empirical quantiles of r^* over the last 30 horizons serve as a non-learned alternative.

For the allocation, let $X_i \sim F_i$ be the forecast of r_i^* and $w_i = w_i^{\mathrm{p}}$. (A class whose deadline exceeds the horizon gets its weight scaled by H/D_i, since a deferred bit may still be served later; this does not apply to the default classes.) We maximise the weighted expected served need,

$$\max_{r} \ \textstyle\sum_i w_i \, \mathbb{E}[\min(B_i + X_i, r_i)] \quad \text{s.t. (3).} \tag{6}$$

Since $\partial_{r_i} \mathbb{E}[\min(B_i + X_i, r_i)] = 1 - F_i(r_i - B_i)$, the probability that the marginal bit is needed, the objective is separable and concave, and the KKT conditions give, for a multiplier $\lambda \geq 0$ on the link constraint,

$$r_i(\lambda) = \mathrm{clip}\big(B_i + F_i^{-1}(1 - \lambda/w_i),\, 0,\, \mathrm{cap}_i\big), \tag{7}$$

with $r_i(\lambda) = 0$ for $\lambda \geq w_i$. Because $\sum_i r_i(\lambda)$ is non-increasing in λ, the λ^* with $\sum_i r_i(\lambda^*) = C$ is found by bisection (60 steps). If $\sum_i r_i(0) \leq C$ the constraint is slack and the surplus is handed out as headroom in proportion to the remaining per-cell room, exactly as in all baselines. The budgets are then held for T slots.

Proposition 1 (Feasibility)**.** *For any forecasts and any λ the budgets* (7) *satisfy the caps, and the returned budgets satisfy $\sum_i r_i \leq C$.*

Proposition 2 (Equal weighted risk)**.** *At an interior optimum $w_i \, (1 - F_i(r_i - B_i)) = \lambda^*$ for every cell: all cells are cut at the same priority-weighted probability of needing one more bit. A wider predictive distribution or a higher weight yields a larger margin above the median. If every F_i is degenerate at $\hat{r}_i$, the rule reduces to a priority-ordered fill: cells receive $B_i + \hat{r}_i$ in decreasing order of w_i, the marginal weight group sharing the remainder in proportion to need.*

Proposition 3 (Backlog first)**.** *Since $F_i(x) = 0$ for $x < 0$, the marginal value of budget below B_i is w_i; every cell with $\lambda^* < w_i$ receives at least $\min(B_i, \mathrm{cap}_i)$ before uncertain demand of any cell of equal or lower weight is funded.*

Proposition 4 (Robustness to miscalibration)**.** *Let G_i be the true distribution of r_i^*, F_i the forecast with $\sup_x |F_i(x) - G_i(x)| \leq \epsilon$ for all i, $J_P(r) = \sum_i w_i \mathbb{E}_{P_i}[\min(B_i + X_i, r_i)]$, and r^F, r^G maximisers of J_F, J_G over* (3)*. Then $J_G(r^G) - J_G(r^F) \leq \epsilon \sum_i w_i [(r_i^G - B_i)^+ + (r_i^F - B_i)^+] \leq 2\epsilon \, w_{\max} \, C$.*

Proof. $\mathbb{E}_P[\min(B_i + X_i, r_i)] = \min(B_i, r_i) + \int_0^{(r_i - B_i)^+} (1 - P_i(x))\, dx$, so $|J_F(r) - J_G(r)| \leq \epsilon \sum_i w_i (r_i - B_i)^+$ for every r. Split $J_G(r^G) - J_G(r^F) = [J_G - J_F](r^G) + [J_F(r^G) - J_F(r^F)] + [J_F - J_G](r^F)$; the middle term is ≤ 0 by optimality of r^F and the others are bounded as above; finally $\sum_i (r_i - B_i)^+ \leq C$. □

The expected weighted unserved need is $\sum_i w_i \mathbb{E}[B_i + X_i] - J$, so the bound also holds for this surrogate of the violation objective; Sec. VII-E tests it empirically.

We also evaluate an online calibration variant that keeps per-cell level offsets $\delta_i \leftarrow \mathrm{clip}(\delta_i + \gamma(\mathbf{1}[y_i > q_i(0.9)] - 0.1), \pm 0.3)$, updated after each horizon as in adaptive conformal inference (ACI) [13]; the rule then uses $F_i^{-1}(p + \delta_i)$.

As for cost, the inverse CDFs take $O(NK)$ and the bisection $O(N n_{\mathrm{iter}})$; the Python prototype needs 3.16 ms per 10-ms epoch on a 4-core CPU, most of it in the 1800 trees, and compiled code takes well under 1 ms.

VI. Evaluation Methodology

Since no public per-slot, per-cell fronthaul traces exist (Sec. II), traffic is simulated with two independent families. Family S is synthetic and bursty. Per cell, user bits are a Gamma-distributed smooth component plus heavy-tailed ON/OFF bursts with Pareto sojourn times [17], modulated by piecewise-constant load regimes (mean 1 s) and converted to fronthaul bits by (2) with a slowly drifting spectral efficiency. ON-state rates stay below each cell's air-interface peak, so violations are caused by the shared link alone. We test five scenarios, with load meaning the mean offered fronthaul load relative to C: low (0.50), moderate (0.65), high (0.80), flash crowds (0.80 with the load multiplied by 1.8 about 15 % of the time, which exceeds capacity) and shift (0.80, with bursts 1.5 times stronger and longer and a heavier tail than anything seen in training or tuning).

Family G is built from standards. Users are dropped in a 3GPP TR 38.901 urban-macro cell (ISD 500 m, 3.5 GHz, 100 MHz, 49 dBm, LOS, path loss and shadowing per the specification), attach to the strongest of seven fully loaded sites and see the other six as interference [19]. Per-slot Rayleigh fading with Jakes correlation at 3 km/h is mapped to spectral efficiency through the TR 36.942 attenuated Shannon bound, quantised to the TS 38.214 CQI table [20], [21]. The eMBB cells carry FTP Model 3 traffic (Poisson 0.5-MByte files, 20 UEs) [22] and the LL cells carry XR video (60 fps, 30 Mbit/s, truncated-Gaussian frame size and jitter) [23]. A per-cell round-robin PRB scheduler serves the UEs, and a cell's fronthaul demand in a slot is its scheduled PRB-layers times β. Only the number of XR UEs per LL cell and the FTP arrival rate are calibrated, so that the mean offered load matches the scenario target (0.65 and 0.80) within 1 %.

Training seeds (1000–1015, loads 0.5–0.95), validation seeds (2000–2003) and test seeds (3000–3019) are disjoint. One forecaster per family is trained once and frozen. Family S: 101,248 rows, validation coverage 5.6/50.7/94.9 % at the 5/50/95 % quantiles, median relative error 37 % against 52 % for persistence. Family G: 101,248 rows, 33.9/56.8/98.1 % and 24 % against 39 %, the 5 % coverage reflecting a point mass at zero. Every tunable parameter of every controller is chosen on the validation seeds (moderate and high scenarios) of its own family. Each test run simulates 20 000 slots (10 s) after a 1000-slot warm-up, every controller seeing the identical trace and stale observations. We report means over 20 seeds and, for paired comparisons, the mean relative reduction with a 95 % percentile-bootstrap interval, the seeds won, and a two-sided Wilcoxon signed-rank test, Holm-corrected across the baselines of a scenario.

Table I lists the controllers, all of which redistribute unused capacity as headroom. Water-fill (WF) gives each cell its need up to its cap if the total fits and otherwise shares the

TABLE I
COMPARED CONTROLLERS (ALL TUNED ON VALIDATION SEEDS ONLY)

Controller	Need estimate / rule	Information
Static equal share	C/N each	none
Reactive proportional	backlog/T + EWMA; WF, unit w	history
MaxWeight [14]	greedy fill by $w_i^{\mathrm{p}} Q_i$ (+ HOL age)	queues
Deadline-aware reactive	B_i + last horizon r^*; WF, w_i	+ deadlines
Point forecast + WF	B_i + forecast median; WF, w_i	+ model
Fixed quantile + WF	$B_i + F_i^{-1}(q)$, one tuned q; WF	+ model
Proposed, window q.	KKT rule, empirical quantiles	history
Proposed (GBM + KKT)	KKT rule, forecast quantiles	+ model
Proposed + ACI	KKT rule, per-cell level offsets	+ model
Proposed, transferred	KKT rule, other family's model	+ model
Oracle water-fill	B_i + true r^*; WF, w_i	future
Oracle + KKT rule	B_i + true r^*; priority fill	future

TABLE II
FAMILY S: V_w (%, MEAN ± STD OVER 20 TEST SEEDS; LOWER IS BETTER; BOLD = BEST IMPLEMENTABLE CONTROLLER)

Controller	Low 0.50	Mod. 0.65	High 0.80	Flash	Shift
Static equal share	4.45±1.05	9.65±1.36	15.46±1.51	19.15±2.34	22.31±2.99
Reactive proportional	0.83±0.27	4.11±0.69	9.20±0.99	12.27±2.23	11.70±2.24
MaxWeight (drift)	0.90±0.28	4.85±0.79	11.54±1.27	15.24±3.02	12.31±2.44
Deadline-aware react.	0.52±0.21	4.24±0.78	11.03±1.33	14.51±3.06	12.56±2.81
Point forecast + WF	0.27±0.17	3.23±0.73	9.70±1.34	13.33±3.09	11.30±2.81
Fixed quantile + WF	0.34±0.23	4.35±1.02	11.76±1.44	15.09±3.17	12.85±3.32
Proposed, window q.	0.25±0.13	2.57±0.64	8.72±1.34	12.67±3.19	10.22±2.88
Proposed (GBM + KKT)	**0.15±0.13**	**2.15±0.60**	**8.00±1.27**	**11.76±3.08**	**10.04±2.71**
Proposed + ACI	0.17±0.13	2.17±0.61	8.14±1.30	12.27±3.26	11.12±3.22
Oracle water-fill	0.07±0.10	1.62±0.56	6.89±1.28	10.16±2.96	6.69±2.60
Oracle + KKT rule	0.05±0.09	0.88±0.42	4.77±1.12	8.10±2.77	5.84±2.37

shortage in proportion to weighted need. MaxWeight solves $\max \sum_i w_i^{\mathrm{p}} Q_i r_i$ subject to (3) and an EWMA usable-rate cap, with a tuned head-of-line-age factor. The fixed-quantile controller reserves one tuned forecast quantile for every cell (level 0.75 in Family S and 0.75 in Family G). Two references use the true r^* of the coming horizon: the oracle water-fill, a perfect point forecast under the baseline rule, and the oracle + KKT rule, the proposed rule with a perfect, degenerate forecast, i.e. the priority-ordered fill of Proposition 2; the gap between the proposed rule and the latter is the cost of forecast error. The transferred variant uses the forecaster trained on the other family.

VII. RESULTS

A. Family S: main comparison

Fig. 1(a) and Table II report V_w per scenario. The proposed rule is the best implementable controller in every scenario. Every paired reduction against a baseline is significant after Holm correction ($p < 0.01$), and the rule wins 20/20 seeds against every baseline in every scenario, except against reactive proportional under flash crowds (4 % [0, 8], 16/20 seeds, $p < 0.05$). The most telling comparison is with the point-forecast controller, which uses the same model, weights and information and differs only in using the median instead of the distribution. Here the reduction is 11–43 % (95 % bootstrap intervals [33, 52] at low load, [10, 12] under shift). Against the tuned deadline-aware reactive controller the reduction is 19–70 %. The tuned fixed-quantile controller is beaten everywhere, because one global percentile over-reserves confident cells and under-reserves uncertain ones. The oracle + KKT reference shows what a perfect forecast would add to the same rule: 40 % of the proposed rule's V_w at high load and 42 % under shift is forecast error, and the remainder is the cost of committing budgets for 10 ms. Fig. 2 shows the mechanism on one LL cell: the proposed rule holds the smallest mean budget of the three deadline-aware controllers (1.62 against 1.66 and 1.72 Mbit per slot) and still loses the fewest bits, because it raises the budget above the median only in epochs whose forecast distribution is wide.

The gain has a clear source. At high load the deadline-unaware controllers keep eMBB nearly clean (4.0 % violations for reactive proportional) but let the LL cells lose 14.7 % of their bits. All deadline-aware controllers protect the LL cells (3.4 % for the proposed rule) and pay for it with eMBB; within that group the proposed rule has the lowest eMBB loss (12.3 % against 16.0 % for the point forecast) and the highest utilisation (72.2 %).

The ablations separate the parts: the KKT rule fed with empirical window quantiles already beats every baseline; the learned model adds 2 % under shift (16/20 seeds) and up to 37 % in the nominal family, so most of the gain comes from the demand notion and the rule rather than from the forecaster. Online conformal calibration makes things worse, by 2 % at high load and 10 % under shift: in a coupled allocation LL misses are partly caused by the shortage split rather than by forecast bias, so the per-cell correction inflates LL reservations and starves eMBB.

B. Family G: standards-based traffic and transfer

Fig. 1(b) and Table III repeat the comparison on the 3GPP-model family, with its own forecaster and its own tuning. The main finding is reproduced: the proposed rule is the best implementable controller at both loads and wins 20/20 seeds against every baseline at load 0.80, with a reduction of 11 % against the point-forecast controller, 31 % against the deadline-aware reactive controller and 50 % against reactive proportional. The traffic is hard in a different way. XR frames arrive every 16.7 ms and each 0.5-MByte FTP file occupies the whole cell for about 13 slots, so the eMBB class loses 20.9 % of its bits even under the oracle + KKT rule with 10-ms budgets, and the deadline-unaware controllers fall behind even static sharing. Unlike in Family S, the proposed rule is here the best implementable controller also under the unweighted ratio (14.4 % against 15.0 % for reactive proportional), because the LL cells are small and periodic enough to be protected almost for free. The transferred variant, whose forecaster has never seen a 3GPP trace, under-reserves the LL cells (5.3 % against 1.3 %) and ends 19 % above the in-family model, a usable allocation but no better than an in-family point forecast. The gap to the oracle + KKT rule is 24 % at high load.

C. The objective matters: unweighted ratio and priority weight

The gain of every deadline-aware controller is a priority trade: fewer LL bits lost, more eMBB bits lost. Under the unweighted violation ratio the proposed rule remains the best implementable controller in only 1 of the five Family S

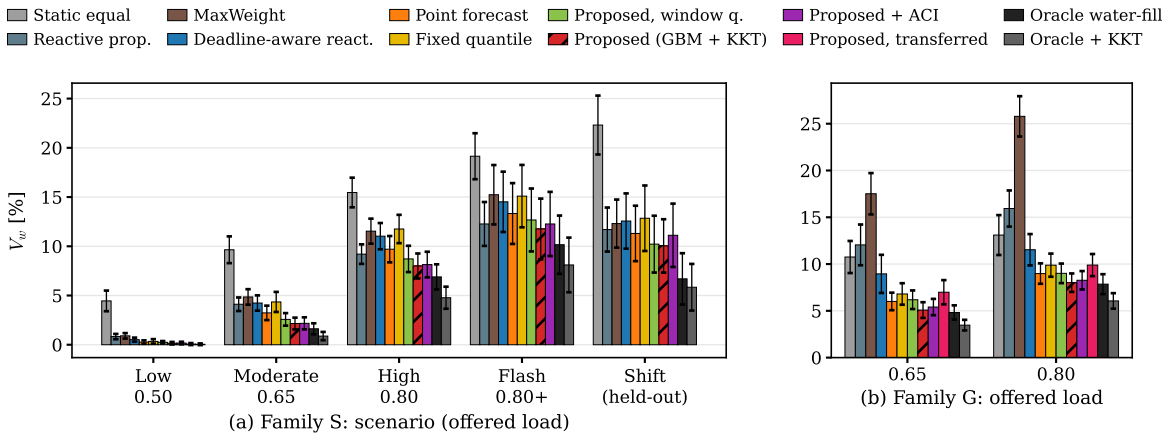


Fig. 1. Priority-weighted deadline-violation ratio V_w (mean ± std over 20 test seeds; lower is better) of all controllers in Table I. (a) Family S, five scenarios; (b) Family G (3GPP TR 38.901 channel, FTP-3 / XR traffic), two loads. Hatched: proposed rule; black and dark grey: oracles (not implementable).

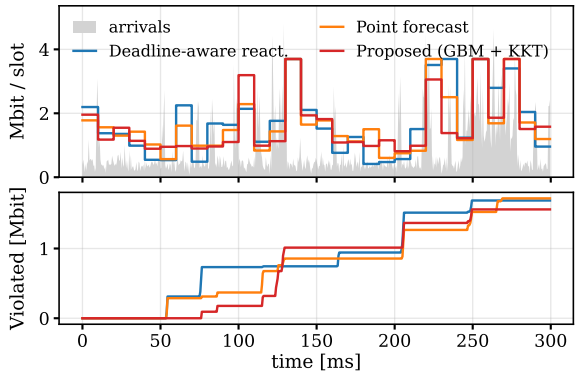


Fig. 2. One low-latency cell of Family S at load 0.80. Top: fronthaul arrivals per slot (grey) and the budget held per 10-ms interval by three deadline-aware controllers. Bottom: cumulative deadline-violated bits.

TABLE III
FAMILY G: V_w (%; MEAN, ± STD AT HIGH LOAD, 20 SEEDS), LL AND UNWEIGHTED RATIO AT HIGH LOAD, AND Δ = PAIRED CHANGE OF V_w TO THE PROPOSED RULE AT HIGH LOAD (%, NEGATIVE FAVOURS PROPOSED; SEEDS WON; */**: HOLM-CORRECTED $p < 0.05/0.01$)

Controller	Mod. 0.65	High 0.80	LL / unw.	Δ (wins)
Static equal share	10.75	13.10±2.13	6.64 / 19.20	-39 (20)**
Reactive proportional	12.05	15.94±1.93	16.96 / 15.01	-50 (20)**
MaxWeight (drift)	17.51	25.79±2.15	32.17 / 19.75	-69 (20)**
Deadline-aware react.	8.95	11.53±1.67	4.24 / 18.47	-31 (20)**
Point forecast + WF	6.00	8.98±1.09	0.98 / 16.59	-11 (20)**
Fixed quantile + WF	6.80	9.88±1.24	0.60 / 18.70	-19 (20)**
Proposed, window q.	6.18	9.01±1.04	1.61 / 16.04	-11 (20)**
Proposed (GBM + KKT)	**5.08**	**8.01±0.99**	1.28 / 14.40	–
Proposed + ACI	5.41	8.27±0.98	1.12 / 15.06	-3 (19)**
Proposed, transferred	6.99	9.89±1.18	5.29 / 14.27	-19 (20)**
Oracle water-fill	4.82	7.85±1.06	0.23 / 15.09	+2 (5)*
Oracle + KKT rule	3.47	6.06±0.83	0.72 / 11.14	+32 (0)**

scenarios (low load). At high load reactive proportional loses fewer total bits (6.5 % against 10.2 %), because it sacrifices the LL cells (14.7 % of their bits) to serve eMBB volume. The weight sweep in the last block of Table IV (every controller is told the new w^{P}, parameters stay tuned at $w_{\mathrm{LL}} = 3$) locates the crossover. In Family S at high load the proposed rule, given $w_{\mathrm{LL}} = 1$, merely matches proportional sharing (6.58 against 6.55 %), and beats it by 13 % at $w_{\mathrm{LL}} = 3$ and 34 % at 5; with a perfect forecast (oracle + KKT) it would win at every weight, so the loss at low weights is forecast error. In Family G at $w_{\mathrm{LL}} = 1$ the rule reaches 15.92 % against 10.43 % for the point-forecast water-fill. XR and FTP demand is zero-inflated (often no frame or file falls into a horizon), and at a low common critical fractile the newsvendor solution reserves nothing beyond backlog for many cells, an all-or-nothing allocation that a median water-fill avoids (LL loss 21.1 % against 5.3 %). At $w_{\mathrm{LL}} = 2$ the two tie, and from 3 on the proposed rule leads (11 % and 22 % against the point forecast). The method is therefore for operators who value a low-latency bit at least two to three times an eMBB bit.

D. Operating regime: interval, delay, compression

Table IV varies the control interval T and the telemetry delay τ on the high-load scenario of Family S, with forecasters retrained per setting, and the BFP mantissa width on the moderate scenario. At $T = 2$ ms the proposed rule reduces V_w by 55 % relative to even the oracle water-fill, which has a perfect point forecast, so the allocation rule and not only the forecast carries the gain (oracle + KKT reaches 1.91 % against 6.89 %). At $T = 20$ ms, five times the LL deadline, reserving the LL peak rate for the whole interval wastes capacity and proportional sharing is better (-11 %). With delay, all controllers degrade as the state ages and the ordering is preserved, but the margin over proportional sharing shrinks from 16 % at $\tau = 0$ to 2 % at $\tau = 5$ ms. Finally, mantissa widths of 6, 9, 12 and 14 bits rescale β in (1) and move the offered load from 0.44 to 0.99 with the same user traffic. At 0.44 nothing is lost whatever the rule, at 0.65 the proposed rule loses about half as much as the reactive controllers, at 0.86 proportional sharing is slightly better on V_w, and at 0.99 no rule helps. Compression thus selects the load regime; deadline-aware allocation matters between about 0.5 and 0.8 and should hand over to proportional sharing or stronger compression above that.

TABLE IV
SENSITIVITY ON FAMILY S: V_w (%, MEANS OVER 10/10/10/20 SEEDS; BOLD = BEST IMPLEMENTABLE)

Control interval T (high)				
Controller	2 ms	5 ms	10 ms	20 ms
Reactive proportional	3.91	7.42	8.91	**11.80**
Deadline-aware reactive	7.89	10.20	10.78	15.84
Point forecast + WF	6.57	7.53	9.46	14.08
Proposed (GBM + KKT)	**3.07**	**5.07**	**7.71**	13.15
Oracle water-fill	6.89	6.01	6.63	10.83
Oracle + KKT rule	1.91	2.74	4.52	8.14
Telemetry delay τ (high)				
Controller	0 ms	1 ms	2.5 ms	5 ms
Reactive proportional	8.16	8.91	9.86	11.30
Deadline-aware reactive	9.89	10.78	11.90	13.93
Point forecast + WF	8.45	9.46	10.51	12.45
Proposed (GBM + KKT)	**6.85**	**7.71**	**8.77**	**11.07**
BFP width / load (moderate)				
Controller	6b/0.44	9b/0.65	12b/0.86	14b/0.99
Reactive proportional	0.22	3.90	**11.37**	**17.29**
Deadline-aware reactive	0.18	4.02	13.84	20.38
Point forecast + WF	0.15	3.00	12.50	19.16
Proposed (GBM + KKT)	**0.11**	**1.96**	12.01	20.81
LL priority weight w_{LL} (high)				
Controller	1	2	3	5
Reactive proportional	**6.55**	**8.13**	9.20	10.56
MaxWeight (drift)	7.38	9.96	11.54	13.38
Deadline-aware reactive	7.75	10.08	11.03	11.69
Point forecast + WF	6.77	8.90	9.70	10.57
Proposed (GBM + KKT)	6.58	8.35	**8.00**	**6.93**
Oracle water-fill	4.86	5.97	6.89	8.29
Oracle + KKT rule	4.86	5.68	4.77	3.63

E. Robustness to miscalibration

To test Proposition 4 we shift every forecast quantile level by $\epsilon \in \{-0.3, -0.2, \ldots, 0.3\}$ before allocation, which is a Kolmogorov distance of $|\epsilon|$ (10 seeds). Under-reserving ($\epsilon < 0$) turns out to be almost harmless (7.80 % at $\epsilon = -0.3$ against 7.71 % at $\epsilon = 0$ at high load): when the link constraint binds, a common shift of all levels is largely absorbed by λ^*, which equalises weighted levels rather than using them absolutely. Over-reserving degrades gradually, to 9.45 % at $\epsilon = +0.3$ (23 % relative), because the upper quantiles saturate at the last knot and the rule loses the tail resolution that tells cells apart. Even then the rule matches the point-forecast controller (9.46 %) and beats the deadline-aware reactive one (10.78 %); the degradation is bounded and roughly linear in ϵ, as predicted.

VIII. DISCUSSION, LIMITATIONS AND CONCLUSION

When budgets must be committed before the traffic is known, turning a forecast *distribution* of deadline-feasible demand into per-cell budgets through the KKT rule reduces priority-weighted deadline violations compared with point forecasts, tuned global quantiles, queue-based and deadline-aware reactive controllers. This holds on two unrelated traffic families and under distribution shift, cross-family transfer and forecast miscalibration. The claim comes with a stated regime: the objective must value the low-latency class (at $w_{\mathrm{LL}} \leq 2$ the simpler controllers are as good or better), control interval and telemetry age must stay within a few times the shortest deadline, above a load of about 0.85 no allocation rule helps and compression or capacity must, and per-stream conformal calibration harms the coupled allocation. Limitations: both families are simulated; Family G uses a round-robin scheduler, flat fading and a fixed rank; the link is a budget-enforced pipe without a switch-queue model; each cell carries one class; HARQ, uplink and the signal-quality cost of compression are not modelled. The next steps are validation on measured per-cell PRB traces and a joint controller for compression and allocation.

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